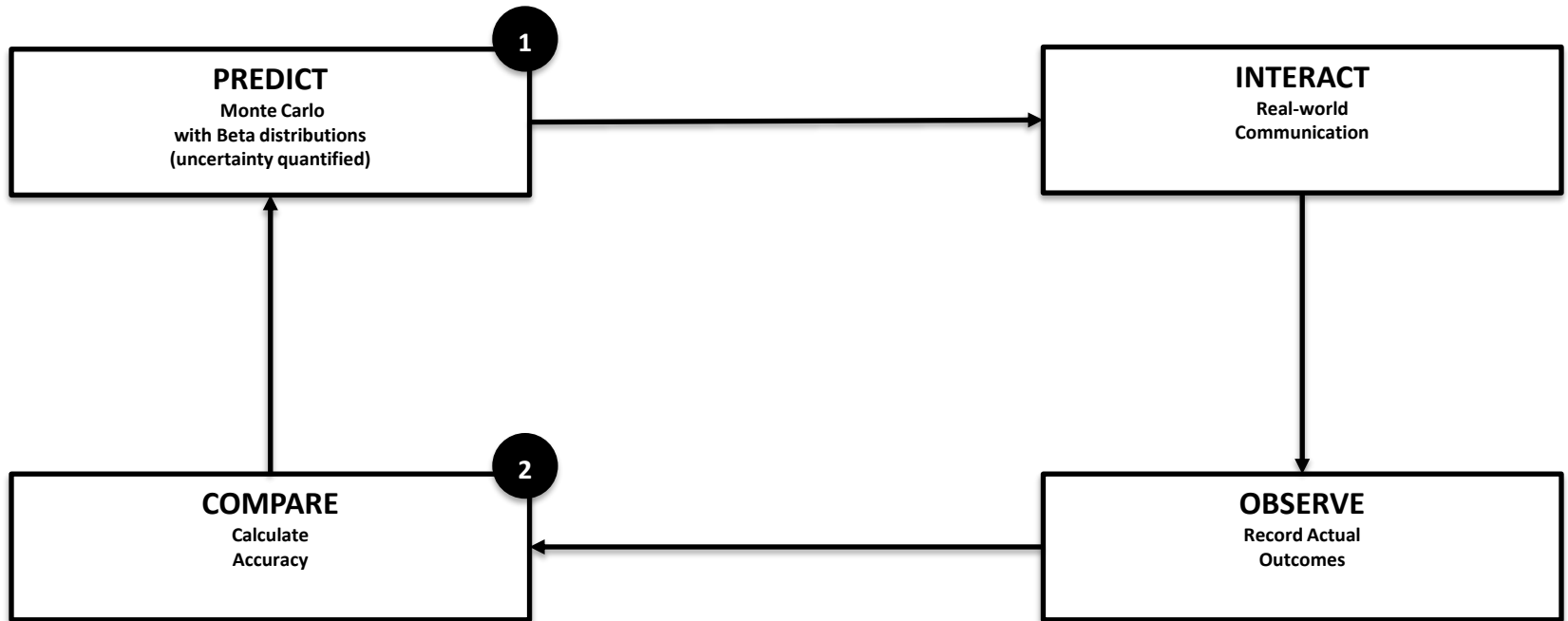


Figure 1: Four-Stage Predictive Learning Cycle



INNOVATIONS:

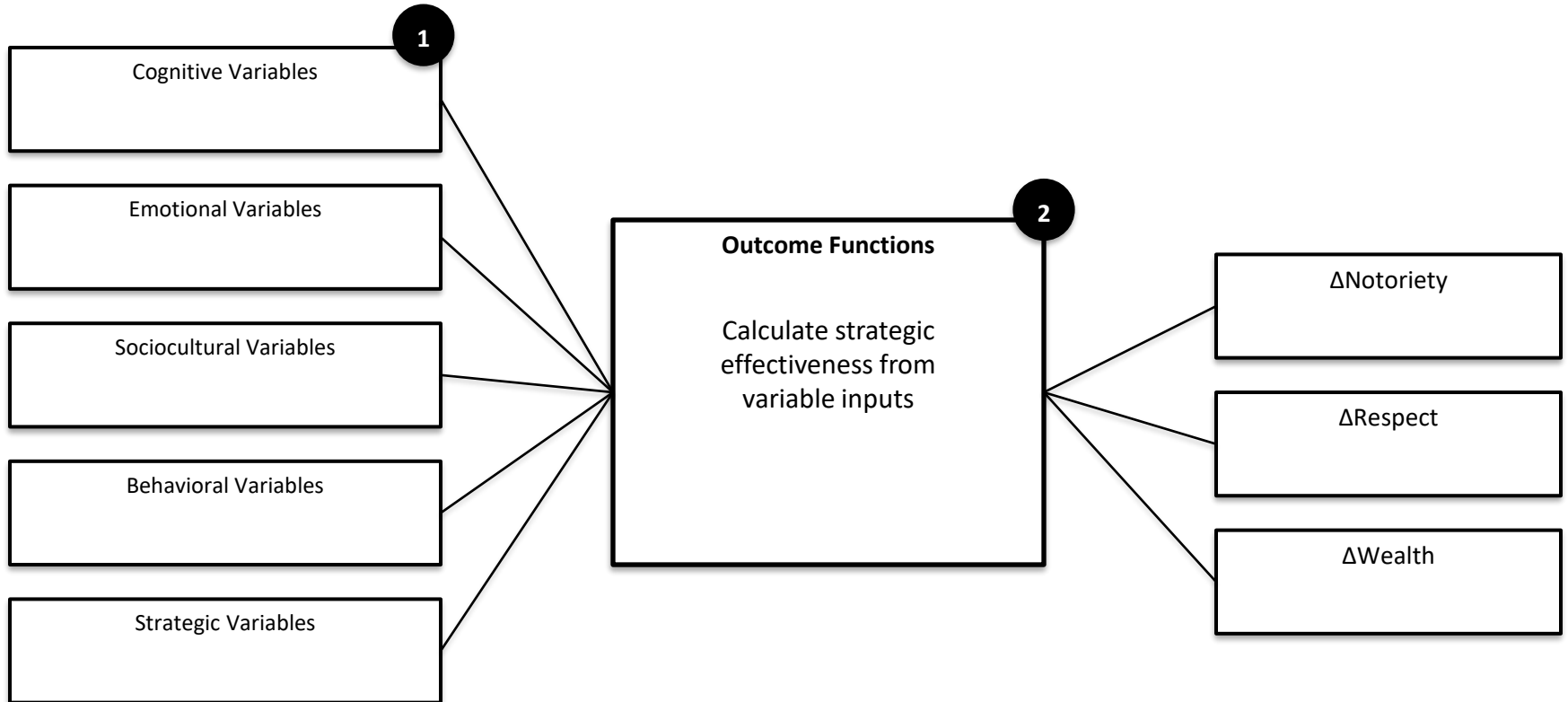
① Monte Carlo Simulation

Novel probabilistic prediction with Beta distributions for uncertainty quantification. (Claims 1, 7, 14)

② Closed-Loop Learning

Systematic comparison of predictions vs. actual outcomes enables bias detection and model improvement. (Claims 1, 11, 24, 25)

Figure 2: Outcome-Driven Variable Ontology



INNOVATIONS:

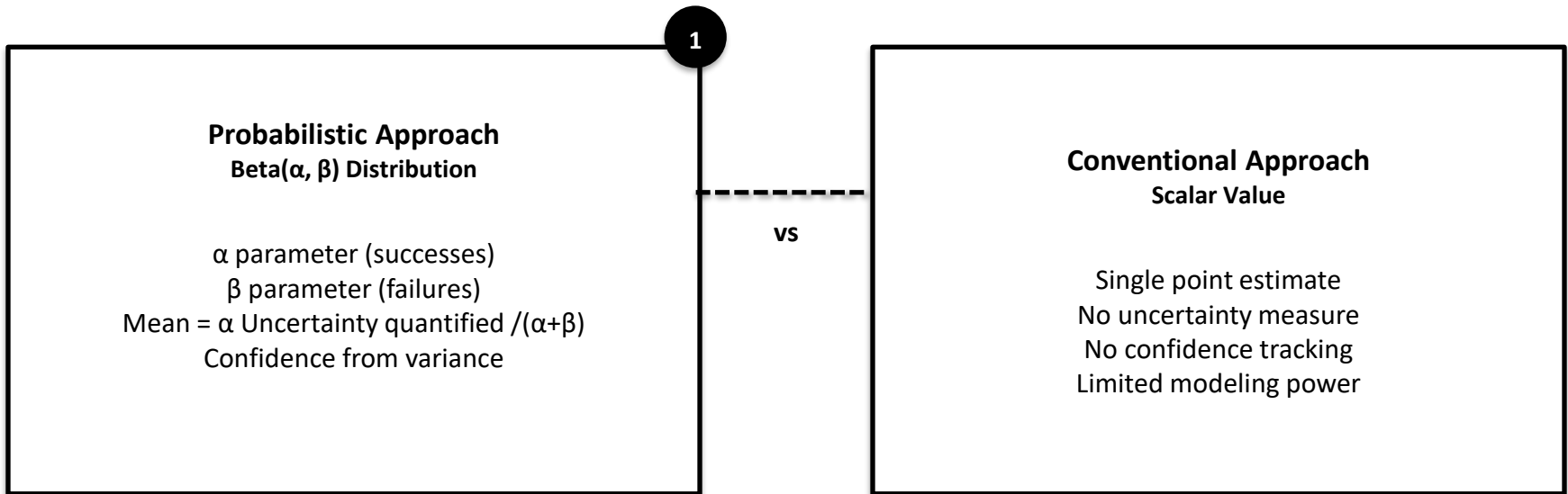
① Five-Layer Ontology

Comprehensive actor modeling across cognitive (8 vars), emotional (7 vars), sociocultural (6 vars), behavioral (7 vars), and strategic (6 vars) dimensions. (Claims 1, 6)

② Outcome Functions

Novel formulas calculate Δ Notoriety, Δ Respect, Δ Wealth from multi-dimensional actor variables. (Claims 1, 21)

Figure 3: Beta Distribution Modeling

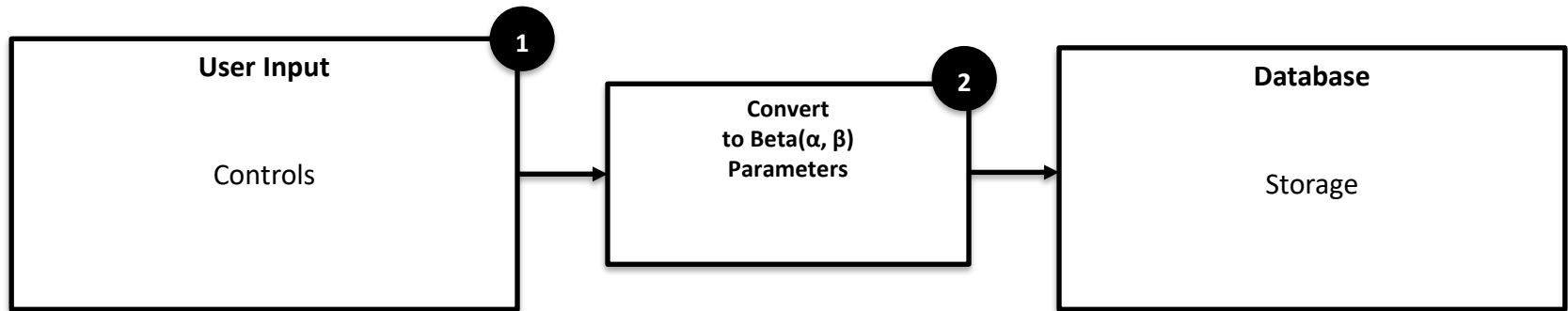


INNOVATIONS:

① Beta Distribution Innovation

Variables stored as Beta(α , β) with parameters for uncertainty quantification, confidence intervals, and Bayesian updating from empirical observations. (Claims 1, 10, 14, 22)

Figure 4: Assessment Interface with Beta Parameter Conversion



INNOVATIONS:

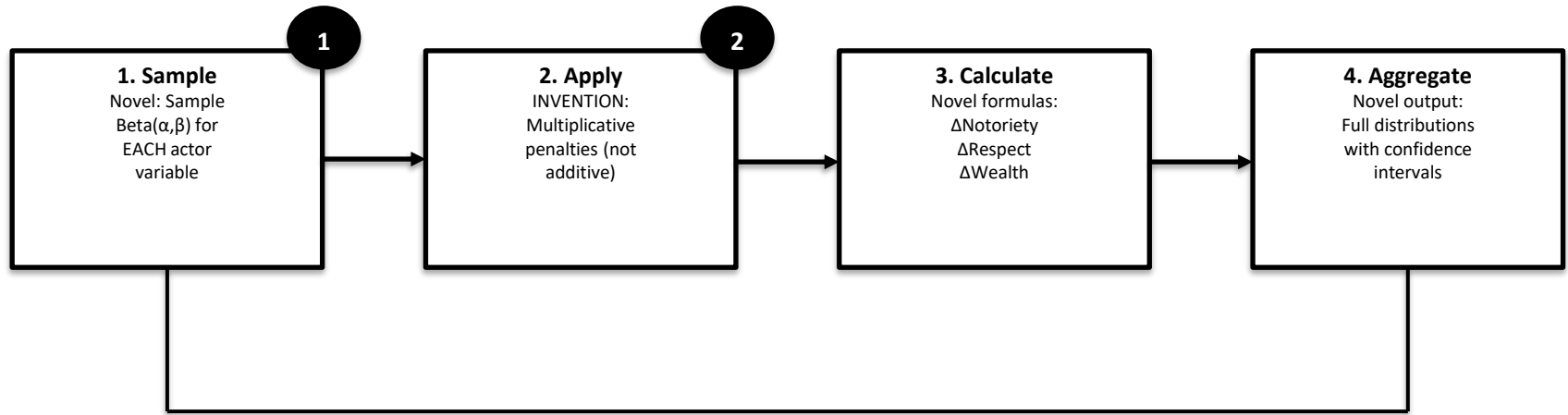
① Intuitive Interface

Users provide estimate (0-100) and confidence (low/med/high) in natural language. (Claims 1, 10)

② Automatic Conversion

System converts intuitive inputs to rigorous Beta(α , β) parameters enabling probabilistic modeling. (Claims 1, 10, 22)

Figure 5: Monte Carlo Simulation Engine



N iterations

INNOVATIONS:

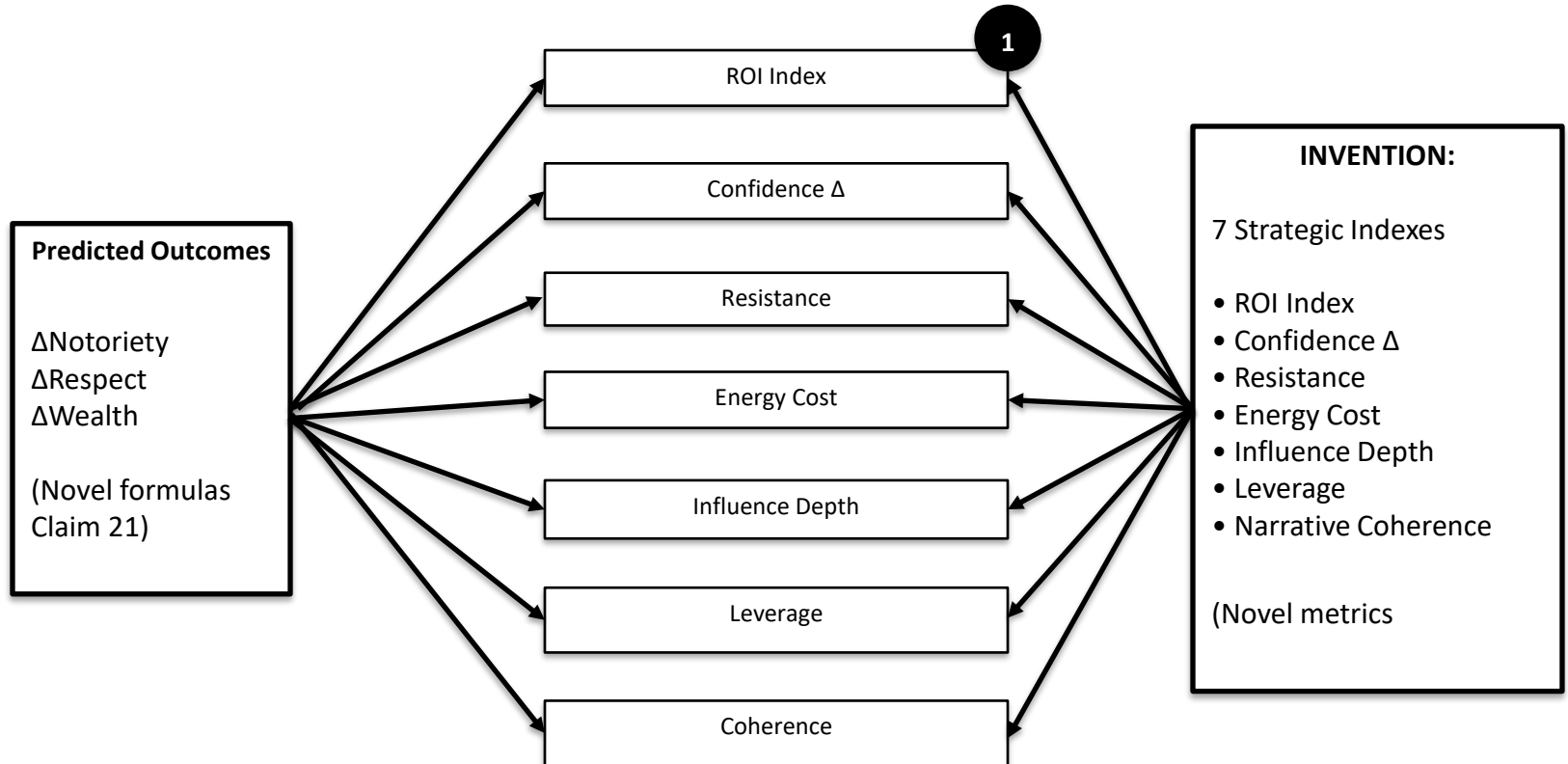
① Beta Distribution Sampling

Each iteration samples from $\text{Beta}(\alpha, \beta)$ for all actor variables, generating probabilistic predictions. (Claims 1, 7, 14)

② Multiplicative Penalties

Novel multiplicative confounder penalties (not additive) applied to skills: stress (-30% obs/adapt), fatigue (-20% obs/adapt), emotion (-20% obs/-30% aware), overconfidence (-40% aware), ego (-30% adapt). (Claims 1, 8, 15, 19)

Figure 6: Strategic Indexes Calculation

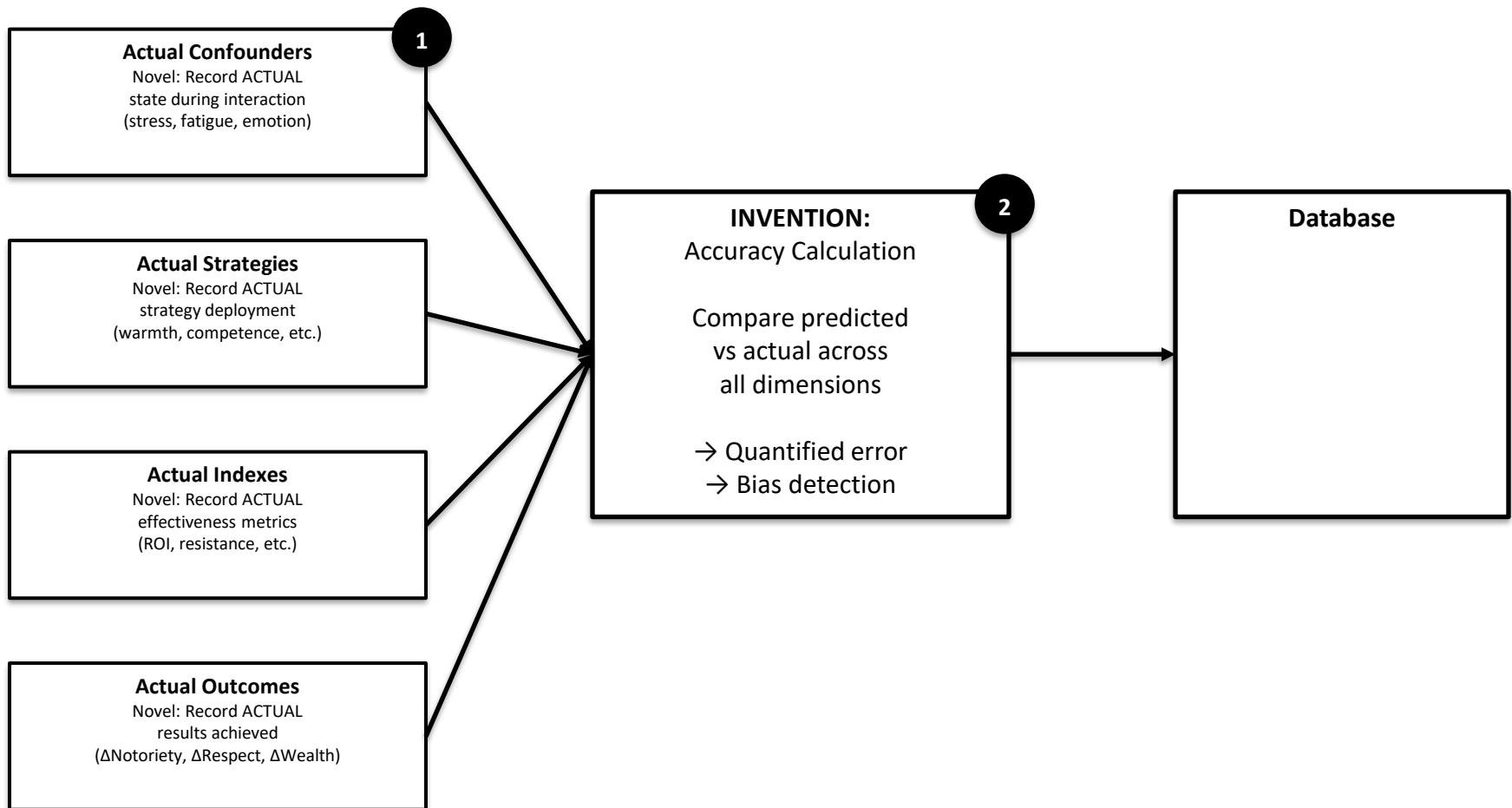


INNOVATIONS:

① Seven-Dimensional Metrics

Novel multi-faceted effectiveness measurement: ROI Index (weighted outcomes), Confidence Δ (estimate certainty change), Resistance Index (friction), Energy Cost (cognitive/emotional load), Influence Depth (belief change), Leverage Activation (advantage creation), Narrative Coherence (message clarity). (Claims 1, 9, 20)

Figure 7: Post-Interaction Recording Interface



INNOVATIONS:

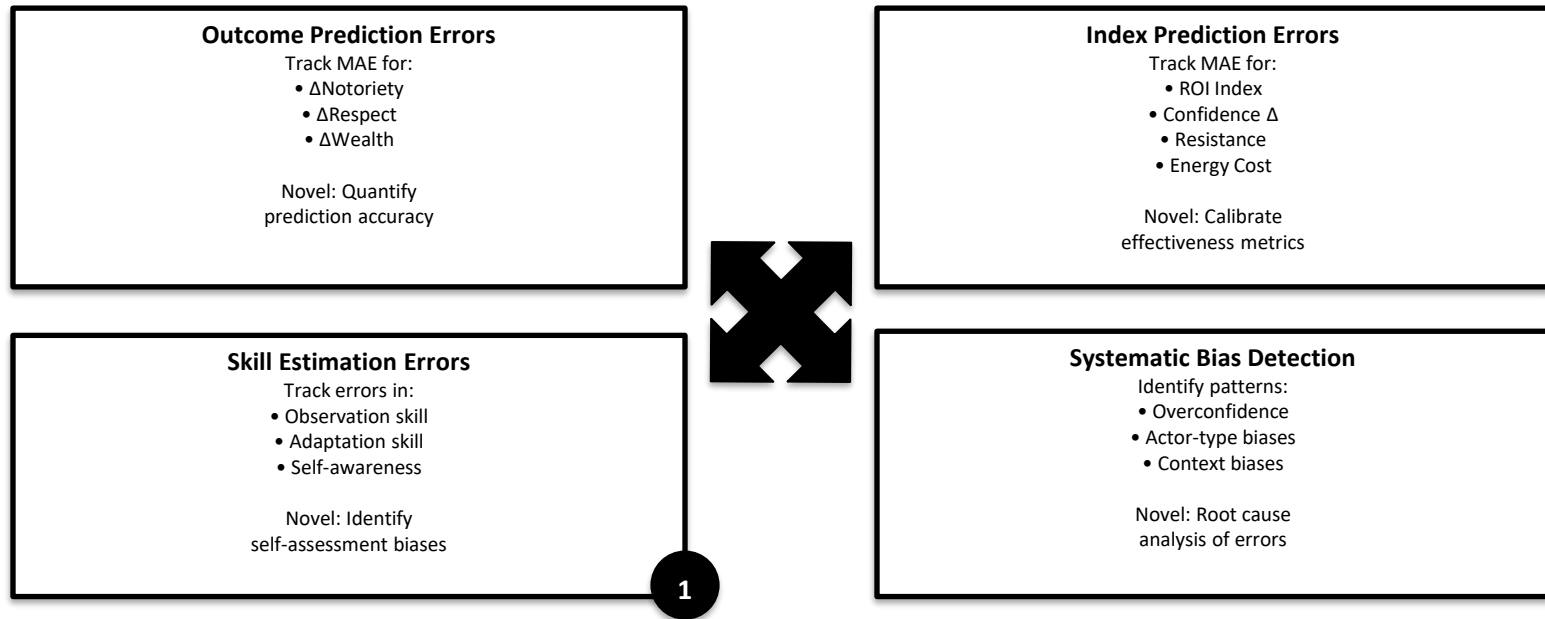
① Systematic Capture

Novel post-interaction recording of ACTUAL confounders, strategies, indexes, and outcomes. (Claims 1, 11, 23)

② Accuracy Calculation

System compares predicted vs. actual values across all dimensions to quantify prediction error and identify systematic biases. (Claims 1, 11, 24, 25)

Figure 8: Accuracy Analysis Dashboard

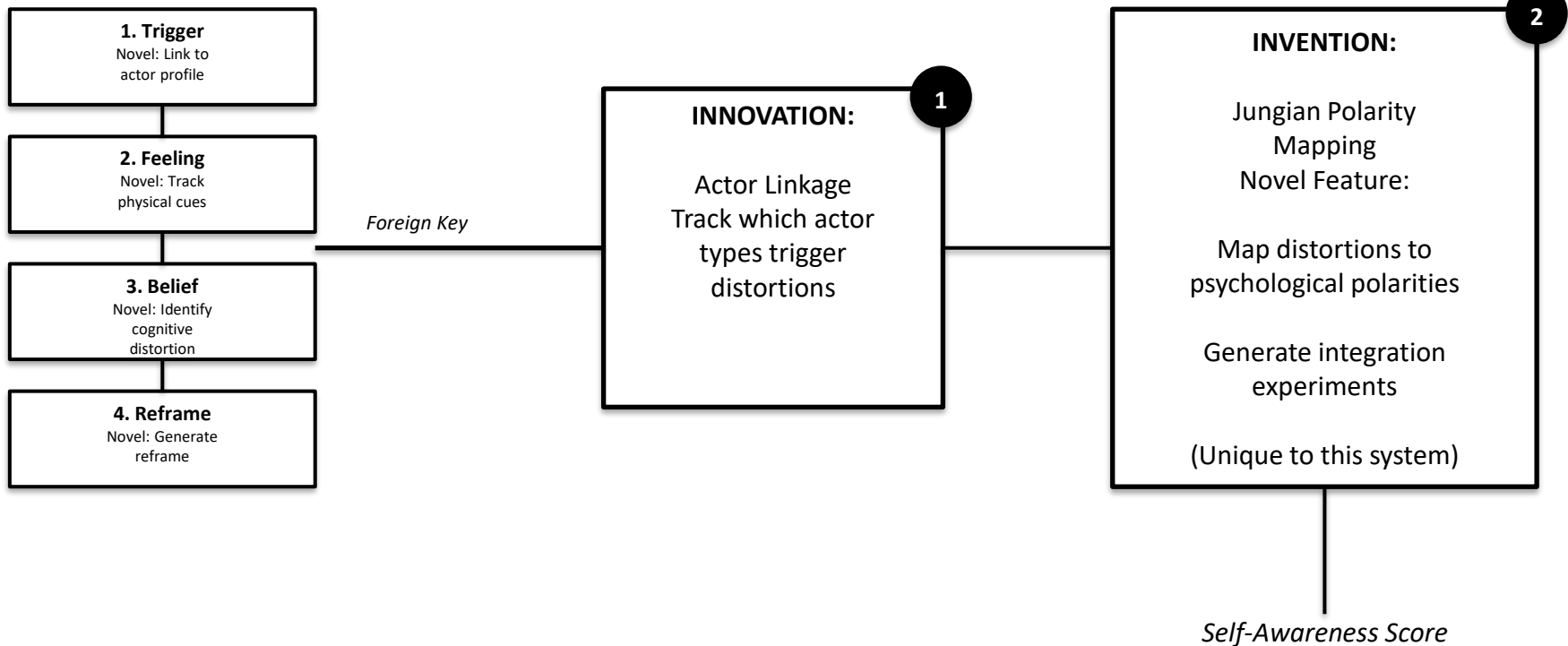


INNOVATIONS:

① Comprehensive Tracking

Novel multi-dimensional accuracy tracking across outcomes (Δ Notoriety/Respect/Wealth), indexes (7 metrics), skills (observation/adaptation/awareness), and systematic bias patterns (overconfidence, actor-type biases, context biases). Enables closed-loop learning and model improvement. (Claims 1, 11, 24, 25)

Figure 9: Cognitive Journal & Jungian Integration



INNOVATIONS:

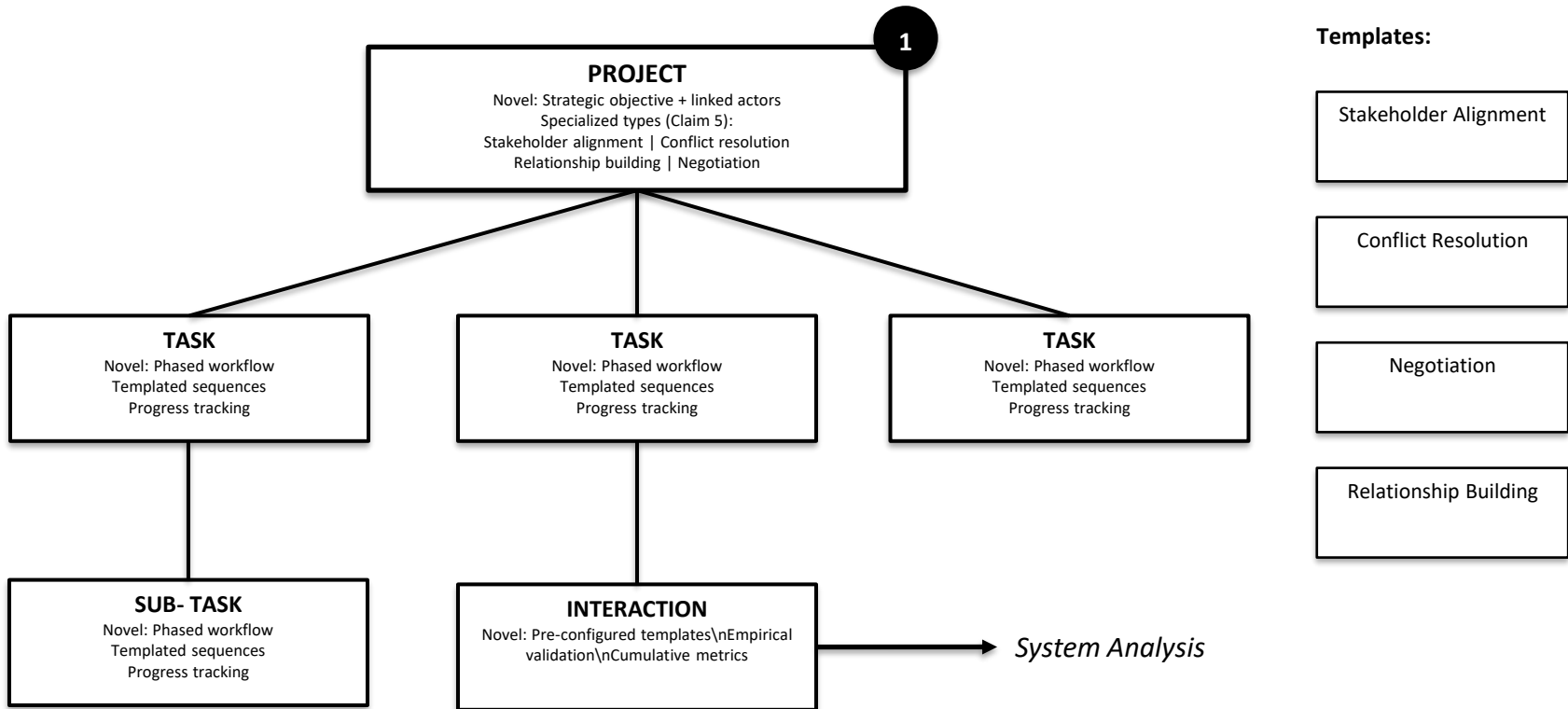
① Actor Linkage

Novel feature linking cognitive distortion episodes to specific actor profiles to identify which actor types trigger distortions. (Claim 2)

② Jungian Integration

Unique psychological mapping to Jungian polarities with generated integration experiments to explore opposite poles. (Claim 3)

Figure 10: Communication Projects Hierarchy

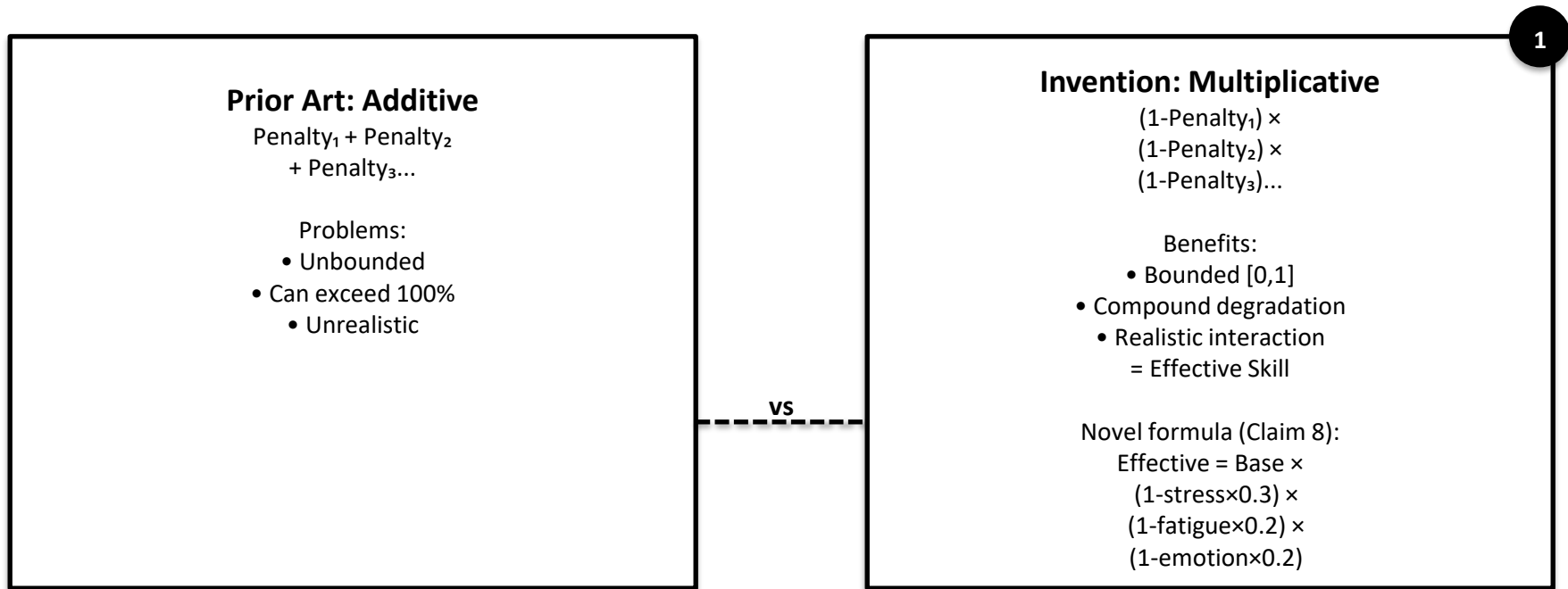


INNOVATIONS:

① Hierarchical Organization

Novel three-level hierarchy (PROJECT → TASK → INTERACTION) with linked actor sets and pre-configured simulation templates. Specialized project types include stakeholder alignment, conflict resolution, relationship building, and negotiation. (Claims 4, 5, 13)

Figure 11: Multiplicative Penalty System



INNOVATIONS:

① Multiplicative Innovation

Novel multiplicative confounder penalties (not additive) ensure bounded [0,1] results with realistic compound degradation. Formula: $\text{Effective} = \text{Base} \times (1 - \text{stress} \times 0.3) \times (1 - \text{fatigue} \times 0.2) \times (1 - \text{emotion} \times 0.2)$. (Claims 1, 8, 15, 19)

Figure 12: Strategy Lever Configuration Interface

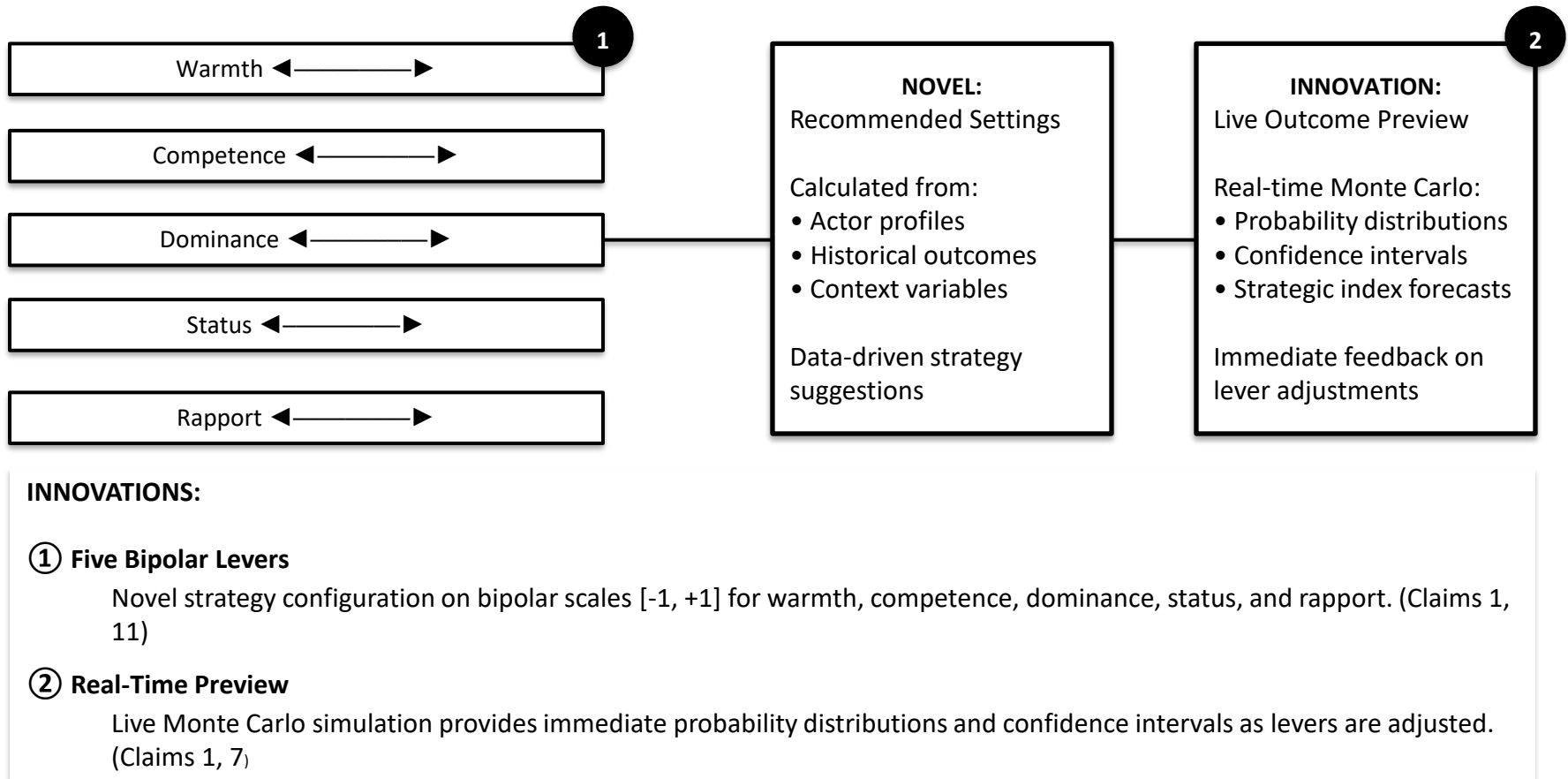
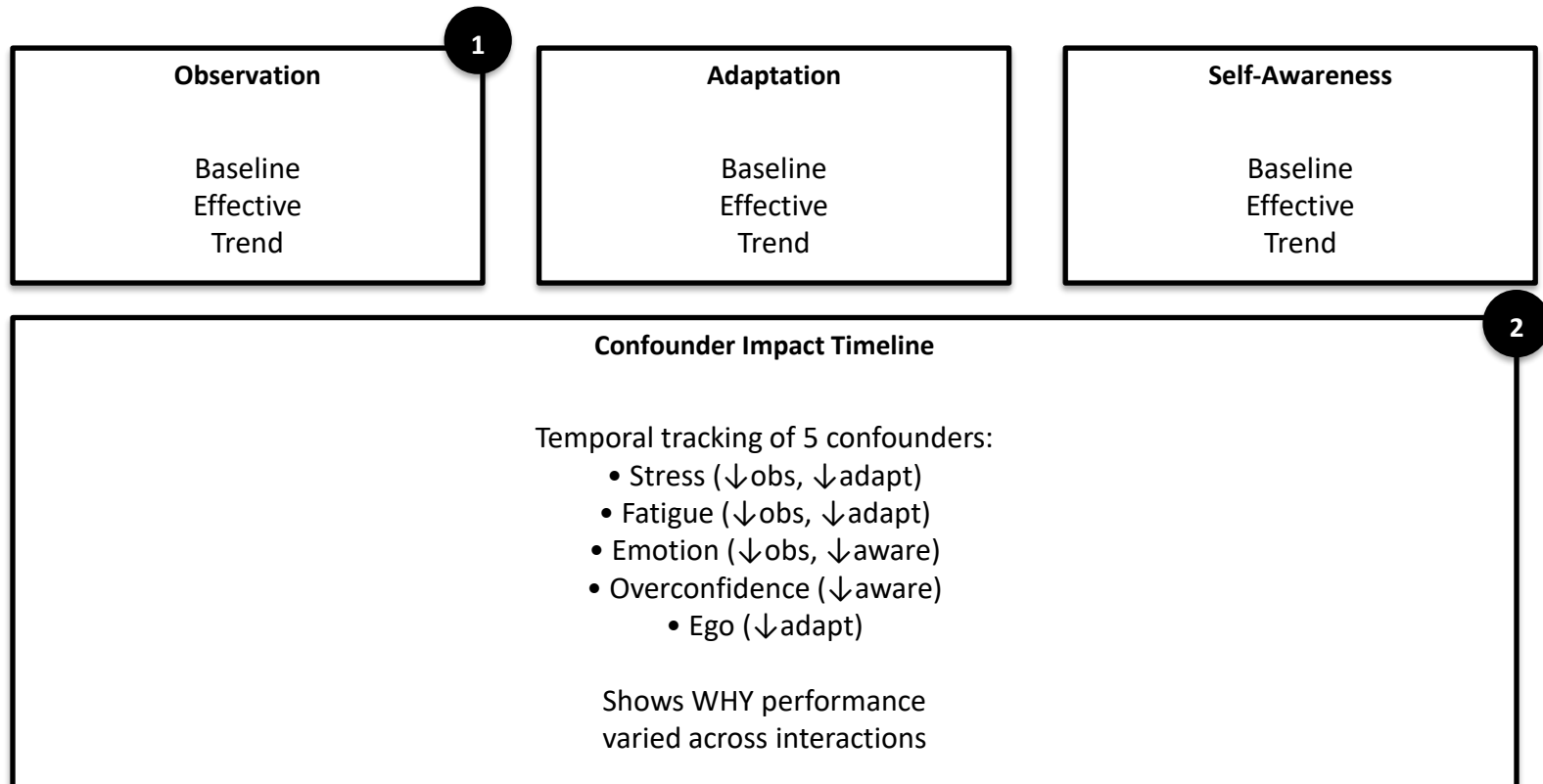


Figure 13: Three-Skill Architect Model with Confounder Tracking



INNOVATIONS:

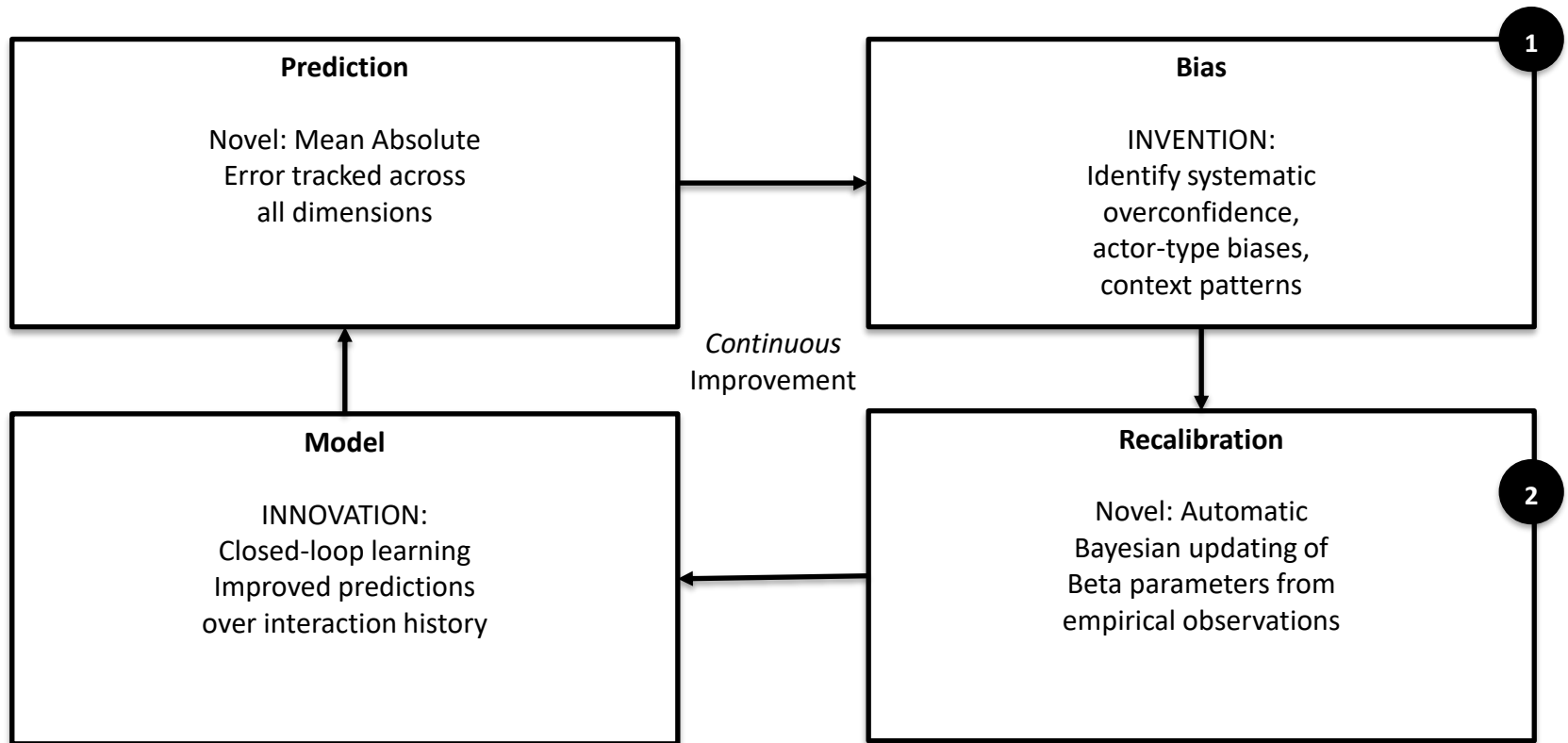
① Three-Skill Model

Novel architect skills tracked as baseline (trait), effective (state-adjusted), and trend (learning curve). (Claims 1, 11, 30)

② Confounder Timeline

Temporal visualization shows when and how 5 confounders degraded skills across interactions. (Claims 1, 30)

Figure 14: Cumulative Learning System



INNOVATIONS:

① Bias Detection

Novel systematic identification of overconfidence, actor-type biases, and context patterns. (Claims 1, 11, 25)

② Bayesian Updating

Automatic recalibration of Beta(α , β) parameters from empirical observations. Closed-loop learning improves predictions over time. (Claims 1, 10, 11, 24)